

Maximum Modulus of Domination Roots of Friendship Graphs: A Sharp First-Order Asymptotic

Xiaojian Zeng

PBC School of Finance, Tsinghua University, Beijing 100190, China

xiaojianzeng@foxmail.com

Abstract

For the friendship graph $F_n = K_1 \vee (nK_2)$, let

$$P_n(z) = (z^2 + 2z)^n + z(1 + z)^{2n}$$

be its domination polynomial, and let R_n be the largest modulus of a zero of P_n . We prove

$$R_n = (\sqrt{2} + o(1)) \sqrt{\frac{n}{\log n}}.$$

The known implicit modulus inequality gives the upper bound after a direct asymptotic argument. For the matching lower bound, we rescale near $z = i\sqrt{2n/\log n}$, formulate an exact equation using principal logarithms on a fixed disk, and apply Rouché's theorem. A uniform error estimate also forces the resulting rescaled zeros to converge to i .

Keywords. domination polynomial; friendship graph; complex zero; Rouché's theorem.

2020 Mathematics Subject Classification. 05C31, 30C15.

1 Statement and context

For a finite graph G , its domination polynomial is

$$D(G, z) = \sum_k d(G, k) z^k,$$

where $d(G, k)$ is the number of dominating vertex sets of size k . The friendship graph F_n consists of n triangles sharing one common vertex. If that common vertex is selected, the peripheral vertices can be chosen arbitrarily; if it is not selected, at least one endpoint of every peripheral edge must be chosen. Hence

$$D(F_n, z) = z(1 + z)^{2n} + (2z + z^2)^n = P_n(z), \tag{1}$$

as obtained in [1, Theorem 2.1].

For every integer $n \geq 1$, define

$$R_n = \max\{|z| : z \in \mathbb{C}, P_n(z) = 0\}.$$

The maximum exists because P_n is a nonconstant polynomial. The question of bounding the moduli of friendship-graph domination roots was raised in [1, Question 3.2]. Rather proved the implicit inequality used below, as well as an earlier linear estimate [2, 3]. The finite-plane limiting locus found in [1] does not determine the rate at which the extreme moduli diverge.

Our result is the following exact first-order asymptotic.

Theorem 1. *With $\log$ denoting the natural logarithm,*

$$\lim_{n \rightarrow \infty, n \geq 2} R_n \sqrt{\frac{\log n}{n}} = \sqrt{2}.$$

The limit is along the full sequence of integers.

Relative to the cited friendship-graph results, the additional ingredient is the construction of a polynomial zero of matching modulus for every sufficiently large n . We derive the all-root upper estimate independently, so the proof is self-contained apart from standard complex analysis. Questions about exact certificates for zeros of other combinatorial polynomials provide broader adjacent context; for example, [4] treats a different polynomial family and supplies no input to Theorem 1. The literature comparison here is limited to the works cited and is not a claim of global priority.

2 A bound for every zero

The elementary estimate in this section is included both for completeness and to make clear that the upper bound covers all complex zeros, not merely a distinguished branch.

Lemma 2. *If $P_n(z) = 0$ and $r = |z| > 1$, then*

$$(r - 1)^2 \log r \leq n. \tag{2}$$

Consequently,

$$\limsup_{n \rightarrow \infty} R_n \sqrt{\frac{\log n}{n}} \leq \sqrt{2}.$$

Proof. We have $P_n(-1) = (-1)^n \neq 0$. For $z \neq -1$, put

$$q(z) = 1 - \frac{1}{(1+z)^2}.$$

Since $z^2 + 2z = (1+z)^2 - 1$, the exact identity

$$P_n(z) = (1+z)^{2n} (q(z)^n + z) \tag{3}$$

shows that every zero satisfies $q(z)^n = -z$. Thus, when $r > 1$,

$$r^{1/n} = |q(z)| \leq 1 + \frac{1}{|1+z|^2} \leq 1 + \frac{1}{(r-1)^2}.$$

Taking real logarithms and using $\log(1+t) \leq t$ for $t \geq 0$ gives (2).

The function $f(r) = (r-1)^2 \log r$ is strictly increasing on $(1, \infty)$. Fix $c > \sqrt{2}$ and set $b_n = c\sqrt{n/\log n}$. Then $b_n > 1$ eventually and

$$\frac{f(b_n)}{n} = c^2 \left(1 - \frac{1}{b_n}\right)^2 \frac{\log b_n}{\log n} \rightarrow \frac{c^2}{2} > 1.$$

For all sufficiently large n , therefore, (2) and monotonicity imply $r < b_n$ for every zero with $r > 1$; zeros with $r \leq 1$ obey the same eventual bound trivially. Hence the normalized limsup is at most c . Letting $c \downarrow \sqrt{2}$ proves the claim. $\square$

3 A logarithmic equation on a fixed disk

It remains to find actual zeros at the scale forced by Lemma 2. For $n \geq 2$, write

$$L = \log n, \quad a = \sqrt{\frac{2n}{L}}, \quad K = \{w \in \mathbb{C} : |w - i| \leq \frac{1}{4}\}.$$

On K we have

$$\operatorname{Im} w \geq \frac{3}{4}, \quad \frac{3}{4} \leq |w| \leq \frac{5}{4}. \quad (4)$$

Let Log denote the principal logarithm on $\mathbb{C} \setminus (-\infty, 0]$. For all sufficiently large n (in particular, whenever $a \geq 4$), define

$$\begin{aligned} u_n(w) &= (1 + aw)^{-2}, \\ H_n(w) &= a^2 \operatorname{Log}(1 - u_n(w)) - \frac{2}{L} \operatorname{Log}(-aw), \\ H(w) &= -w^{-2} - 1. \end{aligned} \quad (5)$$

These functions are holomorphic on a neighborhood of K . Indeed, (4) yields

$$|u_n(w)| \leq \frac{16}{9a^2} \leq \frac{1}{9}, \quad \operatorname{Re}(1 - u_n(w)) \geq \frac{8}{9},$$

so the first logarithm stays in the right half-plane. Moreover, $\operatorname{Im}(-aw) \leq -3a/4$, so the argument of the second logarithm stays in the lower half-plane. The pole $w = -1/a$, the point $w = 0$, and both logarithmic branch cuts are separated from K . The strict inequalities on the compact set also give the asserted neighborhood.

Lemma 3. *Uniformly for $w \in K$,*

$$|H_n(w) - H(w)| \leq E_n, \quad E_n = \frac{128}{27a} + \frac{16}{9a^2} + \frac{|\log L - \log 2| + 8}{L}, \quad (6)$$

and $E_n \rightarrow 0$.

Proof. For $|u| \leq 1/9$, the convergent Taylor series of the principal logarithm gives

$$|\operatorname{Log}(1 - u) + u| \leq \sum_{k \geq 2} \frac{|u|^k}{k} \leq \frac{|u|^2}{2(1 - |u|)}.$$

The preceding bound on u_n therefore implies

$$a^2 |\operatorname{Log}(1 - u_n) + u_n| \leq \frac{16}{9a^2}. \quad (7)$$

Furthermore, $a^2 u_n(w) = (w + 1/a)^{-2}$. Integrating the derivative $-2(w + s)^{-3}$ along $0 \leq s \leq 1/a$, and using $|w + s| \geq \operatorname{Im}(w + s) \geq 3/4$, gives

$$|(w + 1/a)^{-2} - w^{-2}| \leq \frac{128}{27a}. \quad (8)$$

Because a is positive real and the relevant points remain in the indicated principal-logarithm domain,

$$\operatorname{Log}(-aw) = \log a + \operatorname{Log}(-w).$$

On K , we have

$$|\operatorname{Log}(-w)| \leq |\log |w|| + |\operatorname{Arg}(-w)| \leq \log(4/3) + \pi < 4.$$

Finally, $2 \log a = L + \log 2 - \log L$. Combining these facts with (7) and (8) gives (6). Since $a, L \rightarrow \infty$ and $(\log L)/L \rightarrow 0$, the stated uniform convergence follows. $\square$

4 Rouché construction and completion of the proof

On the boundary $|w - i| = 1/4$, the factorization

$$H(w) = -\frac{(w - i)(w + i)}{w^2}$$

and the geometry of K give

$$|H(w)| \geq \frac{(1/4)(7/4)}{(5/4)^2} = \frac{7}{25}. \quad (9)$$

Inside that circle, H has exactly one zero, the simple zero i ; its other zero $-i$ and its pole 0 lie outside. Lemma 3 shows that $E_n < 7/25$ for every sufficiently large integer n . Rouché's theorem then supplies a zero w_n of H_n in $|w - i| < 1/4$ for every such n .

Membership in this fixed disk alone would not yield the sharp constant, so we record the stronger consequence of the uniform estimate. For every $w \in K$,

$$|H(w)| = |w - i| \frac{|w + i|}{|w|^2} \geq \frac{28}{25} |w - i|.$$

Since $H_n(w_n) = 0$, Lemma 3 implies

$$|w_n - i| \leq \frac{25}{28} E_n \rightarrow 0. \quad (10)$$

In particular, $|w_n| \rightarrow 1$.

We next verify that the logarithmic zero is an exact polynomial zero, rather than merely an asymptotic surrogate. Multiplying $H_n(w_n) = 0$ by $L/2$ and using $a^2 L/2 = n$ gives

$$n \operatorname{Log}(1 - (1 + aw_n)^{-2}) = \operatorname{Log}(-aw_n).$$

Exponentiation is legitimate on the branches fixed above, and n is an integer. Hence

$$[1 - (1 + aw_n)^{-2}]^n = -aw_n.$$

Set $z_n = aw_n$. Its imaginary part is positive, so $z_n \neq -1$; the exact identity (3) now yields $P_n(z_n) = 0$. Notice that we use only this implication. We neither assert that every polynomial zero solves the chosen zero-winding logarithmic equation nor infer a polynomial zero count from the Rouché count.

For every sufficiently large integer n , therefore,

$$R_n \sqrt{\frac{\log n}{n}} \geq |z_n| \sqrt{\frac{\log n}{n}} = \sqrt{2} |w_n| \rightarrow \sqrt{2}.$$

This proves the required liminf. Combining it with Lemma 2 proves Theorem 1.

Scope and reproducibility

The theorem concerns all complex zeros and the full integer sequence. The argument constructs one zero of the required scale for each sufficiently large n but does not identify a finite- n maximizer, classify all extreme zeros, prove uniqueness of a polynomial zero in the rescaled disk, or give a higher-order expansion. None of those stronger assertions is needed for the limit.

AI systems assisted in formulating, organizing, and auditing the proof, including exact checks of rational constants and finite checks of domination polynomial coefficients for $1 \leq n \leq 5$. Those computations are sanity checks only; the infinite theorem rests on the analytic proof above. No formal proof assistant was used, and AI-assisted review is not external peer review.

References

- [1] S. Alikhani, J. I. Brown, and S. Jahari, On the domination polynomials of friendship graphs, *Filomat* **30** (2016), no. 1, 169–178. <https://doi.org/10.2298/FIL1601169A>.
- [2] B. A. Rather, On domination polynomials of some graphs, *Journal of Combinatorial Mathematics and Combinatorial Computing* **126** (2025), 279–289.
- [3] B. A. Rather, On roots of domination polynomials for friendship and book graphs, arXiv:2604.08998v2, 2026. <https://arxiv.org/abs/2604.08998v2>.
- [4] Z. Zeng, An eight-element counterexample to real-rootedness of preorder support polynomials, SSRN working paper 7385158, 2026. <https://doi.org/10.2139/ssrn.7385158>.